

High Power Rocket Projects

Rocketry, Builds, Status: Ongoing

Overview

A growing collection of my high power rocket builds: airframes, recovery systems, and avionics each carried from design and simulation through launch. This is where I log projects beyond any single competition entry.

Certification Rockets

Serpent - TRA Level 1 Certification - April 2025

Built from a LOC 4 kit as part of the MSOE HPRC Level 1 Build Day, where club leadership and members of our local Tripoli chapter helped members build and prep their rockets for a certification flight.



Serpent, ready to fly.

- **Motor:** Aerotech H100
- **Recovery:** Single deploy
- **Apogee:** 500 ft
- **Electronics:** None

An issue with the H100's drilled delay timer caused the deployment charge to fire immediately at motor burnout, resulting in a lower apogee than expected. The rocket still recovered safely, and the TRA Level 1 certification was granted.

Grinch - TRA Level 2 Attempt - February 2026

To field a team at the Argonia Cup, the club needs at least one Level 2 certified flier, so I needed to earn that certification before the competition. To keep costs down, I built Grinch mostly from leftover materials from previous club projects: a 2.8 inch OD G12 fiberglass airframe with a 3D printed carbon fiber nylon nosecone and coupler. To save further, I chose single deploy recovery, which is unconventional for a Level 2 class rocket. The fins were 1/4 inch plywood, tacked to the airframe and reinforced with two layers of fiberglass layup.



Grinch on the pad.

- **Motor:** Aerotech J435
- **Recovery:** Single deploy
- **Apogee:** 4,800 ft
- **Electronics:** AirTag GPS tracking

This flight ultimately failed due to a lapse in design judgment: my flight simulation was flawed and overestimated the deployment time for the separation charge. As a result, the charge fired long after apogee and put the recovery harness under forces it wasn't designed for. The shock cord snapped, separating the upper and lower sections. The upper section recovered safely under parachute, but the lower airframe came down ballistic about 1,000 ft from the pad, burying itself 6 inches into the ground and tearing the motor mount tube loose. While the certification attempt failed, I recovered every part of the rocket and rebuilt it with a 20 ft Kevlar shock cord epoxied directly to the motor mount tube, rather than tied to an eyebolt. Drilling through the wooden centering ring to mount that eyebolt had weakened the harness considerably, and it was the direct cause of the failure under recovery loads higher than it was designed for. I rebuilt Grinch as a backup rocket in case my next certification attempt failed.

Scarab - TRA Level 2 Attempt - March 2026

After Grinch's failure, I needed a rocket that would work without question. That led to an entirely new design built around going "low and slow" to guarantee a safe, reliable certification flight. It used a 4 inch diameter cardboard airframe to increase drag, large wooden fins for stability at low speeds, and a recovery harness design already proven on other members' certification flights.



Scarab on the pad.

- **Motor:** AerotechJ425
- **Recovery:** single deploy
- **Apogee:** 4,000 ft
- **Electronics:** GPSTracking

This flight was a resounding success: the separation charge fired correctly at apogee, the parachute brought Scarab down safely, and I was granted my Level 2 certification. The design worked so well that it's now planned for use in HPRC's proposed Level 2 Build Day. On top of that, it was a low-budget build overall, with the motor as its single largest expense.

Competition Rockets 2025 Argonia Cup Rocket - March 2025

A two-stage competition vehicle built as a structures team member for the club's Two-Stage Rocket Team, designed to carry a 16-golf-ball payload to a target apogee of 16,000 ft for the 2025 Argonia Cup. It featured hot-staging separation and dual deployment recovery on both stages, built around a 3 inch fiberglass airframe with 6 layers of fiberglass layup on all fins.



The 2025 Argonia Cup rocket on the pad.

- **Motor:** J135 sustainer / L850 booster
- **Recovery:** Dual deployment, both stages
- **Apogee:** 5,000 ft
- **Electronics:** EasyMini altimeter (booster), EasyMega (sustainer)

This rocket unfortunately failed due to a design flaw in the electronics bays. To safely power them, we used momentary switches to connect and disconnect the flight computers, but the switches we chose were susceptible to vibration and caused the flight computers to lose power after motor ignition. With no power, none of the separation charges fired: the booster came down ballistically and the sustainer was lost. This failure led to a complete overhaul of the club's e-bay design methodology and reinforced the need for true dual redundancy in recovery electronics.

Argonia Cup Rockets - September 2025 to March 2026

Two-Stage Testbed

A half-scale two-stage rocket built and flown during the fall 2025 semester to prove out the staging and recovery configuration before committing to the full-scale competition vehicle, using the same electronics and recovery layout. The first launch attempt saw the sustainer fail to ignite due to a staging logic error, but the recovery system worked correctly, and the data recovered from that flight let the team fix the issue. The second flight succeeded.



With the testbed rocket on the pad.

- **Motor:** H100 sustainer / H550 booster
- **Recovery:** Dual deployment, both stages
- **Apogee:** 3,003 ft
- **Electronics:** Dual EasyMini altimeters (booster), Fluctus + EggTimer Quasar (sustainer)

2026 Argonia Cup Competition Rocket

The full-scale, two-stage competition vehicle built as Project Manager for the club's Two-Stage Rocket Team, designed to carry an 18-golf-ball payload to a target apogee of 18,000 ft. Full writeup, structural and avionics specs, and photos are on the Two-Stage High Power Rocket project page.



The 2026 Argonia Cup rocket on the pad.

- **Motor:** K270 sustainer / K1100 booster
- **Recovery:** Dual deployment, both stages
- **Apogee:** 14,000 ft
- **Electronics:** Dual EasyMini altimeters (booster), Fluctus + EggTimer Quasar (sustainer)

Placed 6th of 21 teams at the 2026 Argonia Rocket Cup.